

Q8.

Extract G

£75m investment at Cadbury

Mondelēz International and the Unite trade union are celebrating successful negotiations over a £75m modernisation to Cadbury's Birmingham factory. Over 200 people agreed to leave by voluntary redundancy by the end of 2016, with some redundancy packages worth in excess of £100 000. It should boost the Cadbury factory's efficiency when four 'state-of-the-art' production lines replace six outdated ones.

5

The 650 remaining staff will develop extra skills with appropriate training provided, for which there will be a 2% pay increase. The investment should enable Cadbury to meet the growing consumer demand for Dairy Milk, Marvellous Creations and Oreo bars.

The table below shows the forecast returns for the first 5 years following the modernisation project.

10

Year	Net cash flow (£m)	Discount factors at 10% discount rate
0	(75)	1.0
1	20	0.909
2	25	0.826
3	30	0.751
4	22	0.683
5	20	0.621

(Sources: adapted from © Guardian News and Media Limited 2014 and © 2017 Trinity Mirror Midlands)

Using the data in Extract G, the Net Present Value (NPV) and Payback, assess Mondelz International's investment in Cadbury's modernisation.

(Total for question = 12 marks)

Q8.

Question Number	Indicative content	Mark
	Knowledge 2, Application 2, Analysis 4, Evaluation 4 Marks for application and analysis – include up to 6 marks for quantitative skills Quantitative skills assessed: QS6: calculate investment appraisal outcomes and interpret results QS8: use and interpret quantitative and non-quantitative information in order to make decisions QS9: interpret, apply and analyse information in written, graphical and numerical forms Indicative content (quantitative skills) - Payback = <u>Cumulative Cash Flows:</u> Year 1 = 20 Year 2 = 45 Year 3 = 75 = 3 years (workings not required) - NPV = £13.81m (workings not required)	

	<table border="1"> <thead> <tr> <th>Net cash flows £m</th><th>Discount rate</th><th>DCF £m</th></tr> </thead> <tbody> <tr> <td>20</td><td>.909</td><td>18.18</td></tr> <tr> <td>25</td><td>.826</td><td>20.65</td></tr> <tr> <td>30</td><td>.751</td><td>22.53</td></tr> <tr> <td>22</td><td>.683</td><td>15.03</td></tr> <tr> <td>20</td><td>.621</td><td>12.42</td></tr> <tr> <td></td><td>Total</td><td>88.81</td></tr> <tr> <td></td><td>NPV</td><td>13.81</td></tr> </tbody> </table>	Net cash flows £m	Discount rate	DCF £m	20	.909	18.18	25	.826	20.65	30	.751	22.53	22	.683	15.03	20	.621	12.42		Total	88.81		NPV	13.81	
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	- It takes exactly three years to pay back the £75m investment which in business is very favourable; usually capital investments are longer term and take 10 years or more to pay for themselves. In the highly competitive chocolate market, earning back an investment quickly can mean that that any profits which follow can be used for other investments such as new plants in emerging economies - The NPV is positive (plus £13.81m) which given that the discount rates were set at 10%, is very favourable, especially given that training costs need to be afforded and pay increases of 2% are to be paid.																									

	<i>Possible counter-balance:</i> • All investment appraisal methods assume that predicted cashflows will be achieved. Should a new chocolate competitor enter the market or Cadbury production of chocolate suffer from a lack of cocoa beans due to drought in Africa, this may not be the case. • A senior manager at Cadbury may have manipulated the forecast data so as to promote the Mondelez International investment in the Birmingham factory in order to save their career • 10% is a high discount rate when compared to the present based rate of 0.25% (2017) therefore a lower discount rate will give you a higher NPV <i>Possible judgement</i> • Focussing on the financial values of the £75m investment may miss out on the qualitative issues. The pros and cons of the project should be considered within the context of corporate objectives, CSR, market position, long term survival – the latter may be very appropriate given the investment enables better production by 'state of the art' production lines. • Given that Cadbury chocolate is worth £4bn in sales to Mondelez International, rising to £6bn by 2020, and that the chocolate products have enjoyed consumer loyalty for many years, it is worth the investment. Also especially given the growing demand for Dairy Milk, Oreos and Marvellous Creations.	(12)
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Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–2	Isolated elements of knowledge and understanding – recall based. Weak or no relevant application to business examples. Generic assertions may be presented.
Level 2	3–4	Elements of knowledge and understanding, which are applied to the business example. Chains of reasoning are presented, but may be assertions or incomplete. A generic or superficial assessment is presented.
Level 3	5–8	Accurate and thorough knowledge and understanding, supported by relevant and effective use of the business behaviour/context. Analytical perspectives are presented, with developed chains of reasoning, showing cause(s) and/or effect(s). An attempt at an assessment is presented, using quantitative and/or qualitative information, though unlikely to show the significance of competing arguments.
Level 4	9–12	Accurate and thorough knowledge and understanding, supported throughout by relevant and effective use of the business behaviour/context. A coherent and logical chain of reasoning, showing cause(s) and/or effect(s). Assessment is balanced, wide ranging and well contextualised, using quantitative and/or qualitative information, and shows an awareness of competing arguments/factors leading to a supported judgement.

Extract A

VisionSpring

VisionSpring is a social enterprise founded to provide glasses that are affordable for people in developing countries, such as India. These glasses improve job opportunities for many people in rural areas.

VisionSpring supplies local small businesses, that sell its glasses, and also trains the business owners to conduct eye tests in local villages. 5

Each pair of glasses costs VisionSpring £1 to make and deliver, which the small business buys for £2 and sells to customers for £4. As every step of the chain is profitable, the business model is sustainable.

VisionSpring’s current growth strategy is to reinvest its profits to expand the scheme, but it is also considering obtaining loans from established businesses in developing countries. 10

One such business is ITC, a tobacco company in India that supplies the majority of village shops.

(Source: adapted from Pyramid power © The Economist, Jan 11th 2007)

Extract B

Economic impact of buying a pair of VisionSpring glasses

Price	£4
Estimated increased earnings per person, per year	£108
Expected life of a pair of glasses	2 years

(Source: adapted from © VisionSpring 2018)

Extract C

Warby Parker

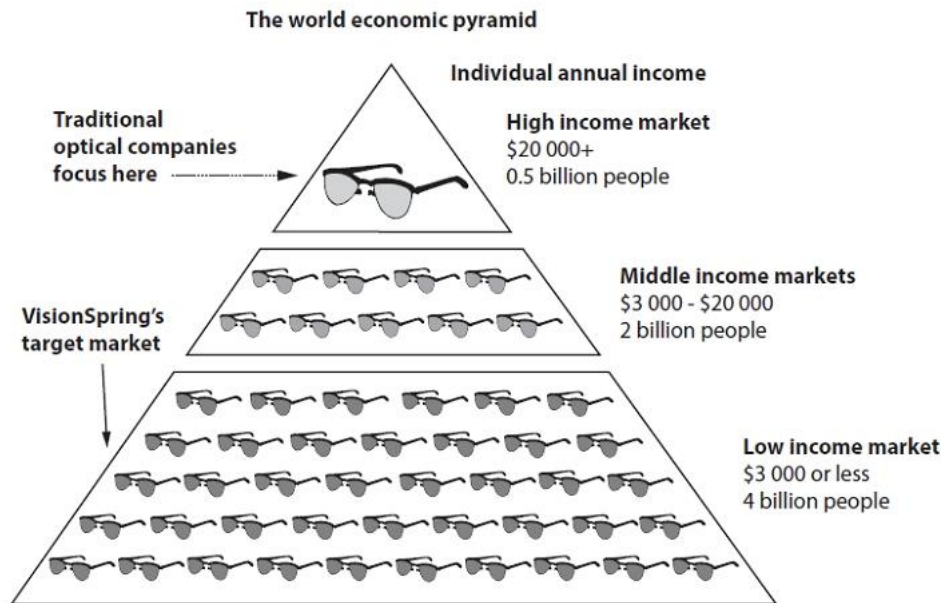
Warby Parker was founded with the objectives of selling designer glasses at a low price and being socially responsible. The optical industry is dominated by a few large companies that keep prices high, making huge profits from consumers who have limited choice. 5

Warby Parker created an alternative by designing and selling glasses to customers directly online. It is able to provide higher-quality, better-looking prescription glasses at a lower price than larger competitors.

Almost one billion people worldwide lack access to glasses, limiting their ability to learn or work. Warby Parker believes that everyone has the right to see. For every pair of glasses it sells another pair is given free to charities, such as Sightsavers, operating in developing countries. 10

(Source: adapted from <https://www.warbyparker.com/history>)

Extract D



In high income markets, glasses are typically sold using a low-volume, high-profit margin approach. However, VisionSpring recognised the potential for success in the developing world. To achieve this, it focused on the low income market, using a high-volume and low-profit margin approach.

(Source: adapted from © VisionSpring 2018)

For people in developing countries, purchasing a pair of VisionSpring glasses may be considered as an investment.

Using the data in Extract B, calculate the average rate of return from purchasing a pair of glasses. You are advised to show your working.

(4)

This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(Total for question = 4 marks)

Q9.

Question Number	Answer	Mark
	Knowledge 1, Application 3 Quantitative skills assessed: QS1: calculate, use and understand ratios, averages and fractions QS2: calculate, use and understand percentages and percentage changes QS6: calculate investment appraisal outcomes and interpret results Knowledge: 1 mark for identifying the formula for calculating Average Rate of Return: average annual return/initial cost x 100 (can be implied) Application: up to 3 marks for • Total return = $216 - 4 = 212$ • Average annual return = $212/2 = 106$ • ARR = $106/4 \times 100 = 2650\%$ NB: if the only answer given is 2650% award 4 marks	(4)

Q10.

Extract E

Buy it Direct

Buy it Direct is one of the largest online retailers in the UK. Established in 1999, it employs over 250 staff across three sites in Huddersfield, Barnsley and the East Midlands. The business is split into separate divisions, each of which is responsible for managing its own budget. The head of each division reports directly to Buy it Direct’s Chief Executive. 5

Buy it Direct was one of the first e-commerce businesses in the UK to sell computers and electronics and then moved into also selling kitchen appliances, televisions and furniture.

Buy it Direct offers competitive prices and high levels of customer service. For example, it has a customer call centre that provides product advice and offers additional services, such as arranging for the installation of new appliances. Buy it Direct aims to hold a wider range of stock than its competitors. It sells premium brands, such as Bosch and Zanussi, as well as lower priced ‘own brand’ products. A significant proportion of its products are imported from China. 10

(Source: adapted from <https://www.laptopsdirect.co.uk/content/buyitdirect>)

Extract F

Buy it Direct’s Chief Executive has set a target of 3 years payback for all investment projects including a new bathroom division with an estimated cost of £700 000.

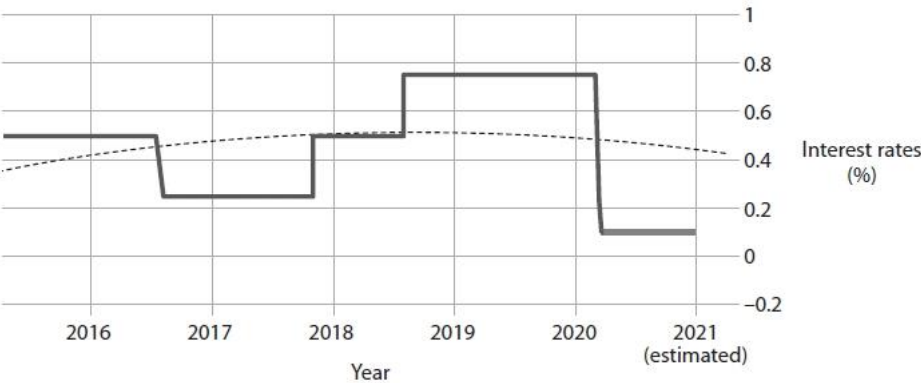
Cash-flow data for Buy it Direct’s proposed new bathroom division

Year	Net cash-flow, £000s
1	200
2	200
3	200
4	200

(Source: figures provided by Nick Glynn, Chief Executive of Buy it Direct)

Extract G

UK interest rates 2016–2021 (estimated forecast for 2021)



(Source: adapted from <https://tradingeconomics.com/united-kingdom/interest-rate>)

Extract H

The UK electrical household appliance sector

It's been a challenging year for the sales of major domestic appliances, such as washing machines and televisions. The market has been growing at just 0.5% as a result of a slowing housing market and wavering consumer confidence. Most major domestic appliance purchases are triggered by replacing a broken or faulty appliance and this part of the market has remained unchanged. When buying household appliances, price and speed of delivery are key factors for consumers. 5

High rates of innovation in the sector and introduction of smart technologies are helping to tempt some consumers to trade up. However, only a small proportion of consumers are willing to pay a higher price for the latest innovations. Meanwhile, research shows that style and design, combined with either price or quality, are the key to deciding which brand of major domestic appliance to buy. 10

(Source: adapted from https://www.retailconomics.co.uk/retail-sector-electricals_market)

The Chief Executive has calculated the new bathroom project will not meet its payback target of three years.

Using the data in Extract F, calculate the difference in the payback period compared to the Chief Executive's target. You are advised to show your working.

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(Total for question = 4 marks)

Q10.

Question Number	Answer	Mark
	Knowledge 1, Application 3 Quantitative skill assessed: QS6: calculate investment appraisal outcomes and interpret results Knowledge: 1 mark for Payback is the length of time a project will take to make the amount of money spent on it Application: up to 3 marks for correct calculations: • Net cash-flow for 3 years is £600,000 (1) • Calculation of payback: 3 years plus $\text{£}100,000/\text{£}200,000 \times 12 = 3 \text{ years and 6 months}$ (1) • The difference is 6 months longer (1) NB: If the only answer given is 6 months award 4 marks.	(4)

Q11.

Before making the investments Jaguar Land Rover would have considered the likely success of the two projects. As well as the payback period and the Average Rate of Return (ARR) they would have considered both the limitations of these techniques and other factors.

Using the information in Extract H and appropriate calculations, evaluate the potential investments at Wolverhampton and Castle Bromwich considering both financial and other factors.

Recommend which project is likely to be the most successful investment for Jaguar Land Rover.

(20)

Q11.

Question Number	Indicative content	Mark
	Knowledge 4, Application 4, Analysis 6, Evaluation 6 - Investment appraisal attempts to determine the value of capital expenditure projects. It enables businesses/investors to compare projects so that a business can expand effectively to best meet its aim and objectives, usually profit maximisation coupled with sales maximisation and efficiency, respectively. - In 2014-15, JLR broke its profits record with a £2.6billion announcement implying that investments by the company more than pay for themselves. Financial factors Payback for the Wolverhampton Project: Year 5 cumulative net cash-flows amount to £470m, leaving £30m to pay for the £500m investment. Year 6 = £120m new cash-flow, therefore this is £10m per month. So 3 months would = £30m. So Payback = total of 5 years and 3 months. This is 4 years 9 months earlier than the Castle Bromwich Project, which pays for itself in 10 years (estimate) - Payback does not, however, take into account the total profit generated over the lifetime of these two projects. Castle Bromwich, over a longer term – say 20 years – could be more profitable for JLR.	
	Average Rate of Return (ARR) for the Wolverhampton Project: - ARR = $\frac{\text{Average profit}}{\text{Cost of initial investment}} \times 100$ Therefore Total Profit = Total Net Cash Flows – Capital Costs/number of years = £1100m - £500m = £600m = £600m/10 = £60m ARR = $\frac{\text{Average profit}}{\text{Cost of initial investment}} \times 100$ $\frac{£60m}{£500m} \times 100$ = 12% - The rate of return of the £500m Land Rover Engine Manufacturing Centre in Wolverhampton is 12%, compared to the £400m Jaguar XF project in Castle Bromwich at 8%. - However, differing initial investment amounts (XF = £400m; Engine Plant £500m) means that the opportunity cost of capital is greater for the Wolverhampton Project. This may have therefore required an extra share issue or more costly borrowings at first, which might reduce the returns on capital	(20)

Other Factors

- Negative externalities caused by each of the investments may differ. There may be more environmental damage/pollution, traffic congestion, disruption to other businesses in the Wolverhampton region compared simply modernising the Castle Bromwich project.
- Investing in the Jaguar XF plant may mean the purchase of high tech robotic equipment to ensure that Jaguar retains its position for quality car production in the prestige car market
- The recruitment and training of 1,400 new employees in Wolverhampton may boost the company's image and in turn sales
- More jobs may be created as a result of the Castle Bromwich project; 1,300 were created in Solihull alone in order to build just the SUV.
- May depend on the objectives and strategies of Jaguar – directors may want to reduce the sub-contracting of engine manufacturing in order to improve quality assurance of all its cars
- The XF investment may mean that Jaguar produces another car which is economic in terms of fuel economy and in terms of CO₂ emissions; significant sales factors given the lower total motoring costs to the consumer

Possible recommendations

- Engines produced in Wolverhampton may be for across the Jaguar range, therefore key to the production and sales of more cars in the longer term. Furthermore the non-financial benefits may prove to be more significant to the company's CSR stance.
- The all new XF may be a cash cow for Jaguar, which could therefore have a more positive immediate impact on sales revenue, cash flow and even market share; vital in a increasingly competitive global market.

NB: to access levels 3 and 4, students will need to demonstrate quantitative skills to support their judgements

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–4	Isolated elements of knowledge and understanding. Weak or no relevant application of business examples. An argument may be attempted, but will be generic and fail to connect cause(s) and/or consequence(s)/effect(s).
Level 2	5–8	Elements of knowledge and understanding, which are applied to the business example. Arguments and chains of reasoning are presented, but connections between cause(s) and/or consequence(s)/ effect(s) are incomplete. Attempts to address the question. A comparison or judgement may be attempted, but it will not successfully show an awareness of the key features of business behaviour or business situation.
Level 3	9–14	Accurate and thorough knowledge and understanding, supported throughout by relevant and effective use of the business behaviour/context. Uses developed chains of reasoning, so that cause(s) and/or consequence(s)/effect(s) are complete, showing an understanding of the question. Arguments are well developed. Quantitative or qualitative information is introduced in an attempt to support judgements, a partial awareness of the validity and/or significance of competing arguments and may lead to a conclusion.
Level 4	15–20	Accurate and thorough knowledge and understanding, supported throughout by use of relevant and effective use of the business behaviour/context. Uses well-developed and logical, coherent chains of reasoning, showing a range of cause(s) and/or effect(s). Arguments are fully developed. Quantitative and/or qualitative information is/are used well to support judgements. A full awareness of the validity and significance of competing arguments/factors, leading to balanced comparisons, judgements and an effective conclusion that proposes a solution and/or recommendations.